

INFECTIOUS DISEASE EPIDEMIOLOGY

INTRODUCTION TO EPIDEMIOLOGICAL MODELS AND DISEASE FORECASTING

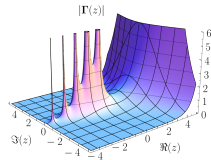
B. K. M. Case

14 September 2022

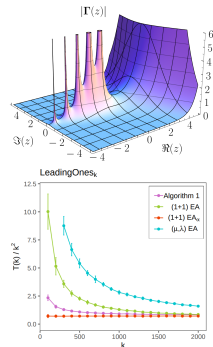
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- Undergraduate in Math at Oberlin College

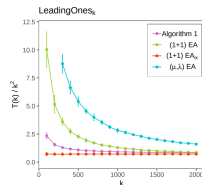
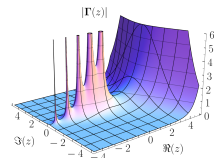


- Undergraduate in Math at Oberlin College
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 - ▶ Studied self-adaption in Evolutionary Algorithms



Theorem 2. Algorithm 1 with $\frac{\lambda}{\mu} = \alpha_0 \geq 4$, constant parameters $A, b, p_{inc} \in \mathbb{R}$ satisfying $A > 1$, $(1+\delta)/\alpha_0 < p_{inc} < 2/5$, and $0 < b < 1/(1+\sqrt{1/(\alpha_0(1-p_{inc})}))$ for some $\delta \in (0, 1)$, parent population size $\mu = \Omega(\log(n))$, and $\lambda \geq c \ln(n)$ for a large enough constant c , for any $k \in \mathbb{N}$, has expected runtime $O(k\lambda \log(n\lambda) + k^2)$ on LEADINGONES_k .

- Undergraduate in Math at Oberlin College
- MRes in Natural Computation at University of Birmingham
 - ▶ Studied self-adaption in Evolutionary Algorithms
- Started with studying complexity science at UVM
 - ▶ Realized I really like applied work in biology and public health!



Why model?

- Find patterns in data, make predictions, find sufficient conditions that replicate data
- Models are required for confidence intervals, p-values, power analyses, etc.

Two types of models

- What is a model? Common distinction between **statistical** vs. **mechanistic** models ¹
- A model is a **latent process** f + **observation process** g

$$\mathbf{y} = g(\boldsymbol{\eta}, e)$$

$$\boldsymbol{\eta} = f(\mathbf{x}, \boldsymbol{\theta})$$

where $\mathbf{y}$ is response data, $\mathbf{x}$ is known input data, $\boldsymbol{\theta}$ are model parameters and e is measurement noise

1. LESSLER et CUMMINGS, "Mechanistic models of infectious disease and their impact on public health", *American Journal of Epidemiology*. 2016.

Model inference : fitting your model to data

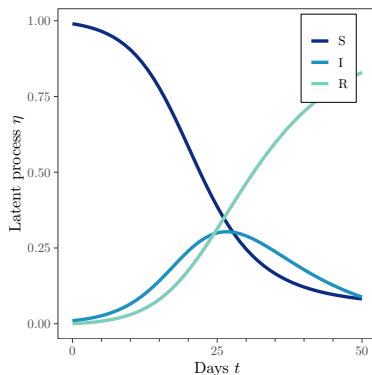
- Models go from **inputs** to **outputs** : $(x, \theta) \mapsto_f \eta \mapsto_g y$
- We are interested in reversing this process :
 - (i) Which input parameters are the *most likely* to have produced y ?
 - (ii) How confident are we these parameters are the "correct" ones?

Initially proposed by Kermack and McKendrick in 1927,² models the proportion of susceptible/infectious/recovered

Model assumptions

- Well mixed. All interactions equally likely
- Compartmental. Individuals in a compartment are identical

$$\begin{aligned}\frac{dS}{dt} &= -\beta SI \\ \frac{dI}{dt} &= \beta SI - \alpha I \\ \frac{dR}{dt} &= \alpha I\end{aligned}$$



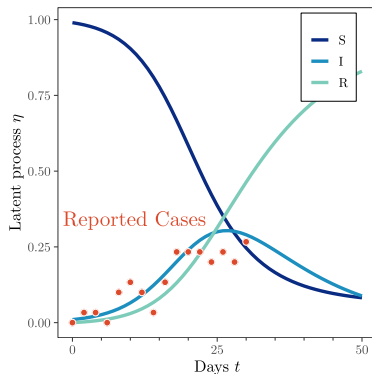
- Induces $\eta(t) = (S(t), I(t), R(t))$

2. KERMACK et MCKENDRICK, "A contribution to the mathematical theory of epidemics", *Proceedings of the Royal Society of London, Series A*. 1927.

Initially proposed by Kermack and McKendrick in 1927,² models the proportion of susceptible/infectious/recovered

Adding observations

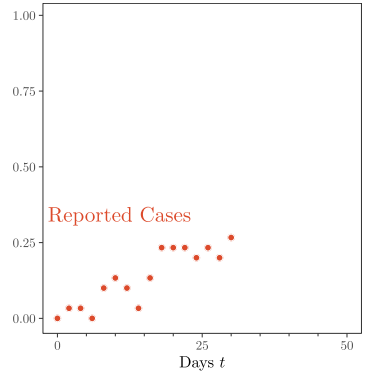
- Suppose we test n people per day, producing daily infections $y_1, y_2, \dots$
- Under unbiased testing, probability a test is positive is $I(t)$



2. KERMACK et MCKENDRICK, "A contribution to the mathematical theory of epidemics", *Proceedings of the Royal Society of London, Series A*. 1927.

The data we see

- WHAT IS THE INFECTION RATE β ?
- HOW MANY PEOPLE ARE INFECTIOUS?
- HOW LONG WILL THE OUTBREAK LAST?



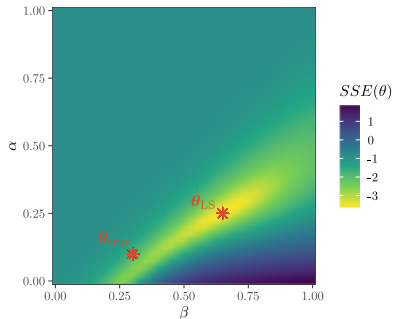
- (i) Which input parameters are the most likely to have produced y ?
- (ii) How confident are we these parameters are the correct ones?

- Need a way to "score" our model parameters, $\theta = (\beta, \alpha)$
- The parameter with the best score is called an **estimator**

Least squares

$$SSE(\theta) = \sum_t \left(\frac{y_t}{n} - I(t; \theta) \right)^2$$

- Penalize larger deviations from data more than small ones
- Estimator is $\hat{\theta}_{LS}$ minimizing the **sum of squared errors**

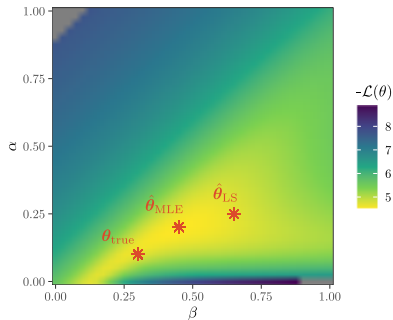


- $\hat{\theta}_{\text{LS}}$ ignores how we collected data (testing)
- The collection process can be written as a **probability distribution** $p(\mathbf{y} \mid \boldsymbol{\theta})$ called a **likelihood**

Maximum likelihood

Maximize $\mathcal{L}$. Assuming testing is independent across days, have

$$\begin{aligned}\mathcal{L}(\boldsymbol{\theta}) &= p(\mathbf{y} \mid \boldsymbol{\theta}) \\ &= \prod_t p(y_t \mid \boldsymbol{\theta}) \\ &= \text{const} \prod_t I(t; \boldsymbol{\theta})^{y_t} (1 - I(t; \boldsymbol{\theta}))^{n - y_t}\end{aligned}$$



- (i) Which input parameters are the most likely to have produced y ?
- (ii) How confident are we these parameters are the correct ones?

Care about more than the "best case" for fitting model to data

- **Frequentist** : estimator $\hat{\theta}$ is drawn from a distribution of hypothetical experiments, repeated many times
- **Bayesian** : no estimator! Our beliefs given data $\mathbf{y}$ are expressed as a **posterior distribution** $p(\theta \mid \mathbf{y})$

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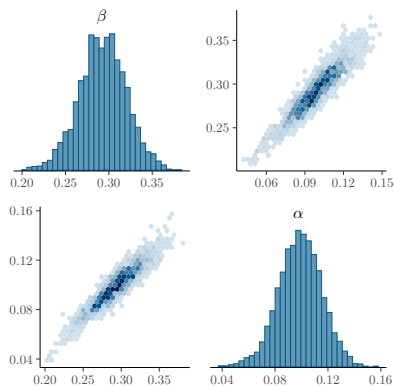
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BAYES' RULE

For data $\mathbf{y}$ and parameters $\boldsymbol{\theta}$,

$$p(\boldsymbol{\theta} \mid \mathbf{y}) = \frac{p(\mathbf{y} \mid \boldsymbol{\theta})p(\boldsymbol{\theta})}{p(\mathbf{y})} \propto \mathcal{L}(\boldsymbol{\theta})p(\boldsymbol{\theta})$$

WHAT IS THE INFECTION RATE β ?

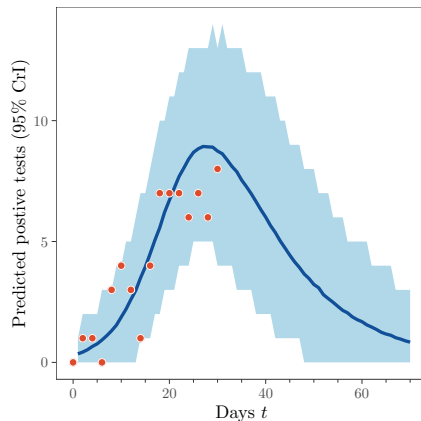


HOW MANY PEOPLE ARE INFECTIOUS?

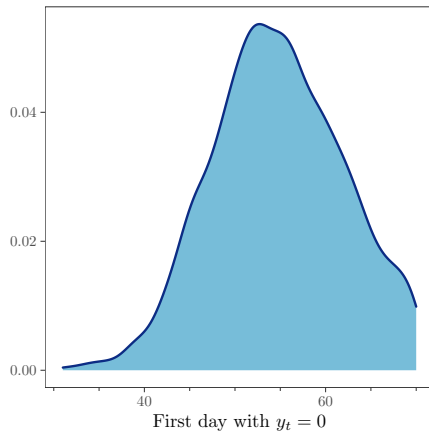
Posterior prediction

Suppose we sample new data $\hat{y}$ by testing more individuals. This has distribution

$$p(\hat{y} \mid \mathbf{y}) = \int p(\hat{y} \mid \mathbf{y}, \boldsymbol{\theta}) p(\boldsymbol{\theta} \mid \mathbf{y}) d\boldsymbol{\theta}$$



HOW LONG WITH THE OUTBREAK LAST?

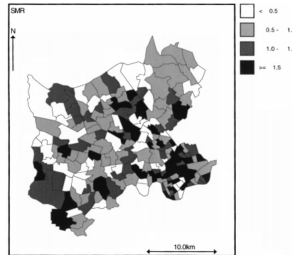


II

HIERARCHICAL MODELS FOR EPIDEMIOLOGICAL DATA

What is epidemiological data?

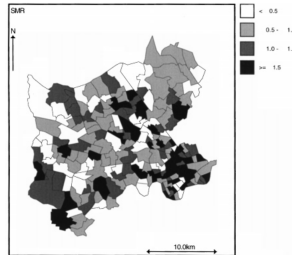
- Commonly measured as **counts**, i.e. positive and no fractions



Relative risk of larynx cancer mortality among men. From Pascutto et al. (2000)

What is epidemiological data ?

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- **Stratified** across time, space, and/or demographic groups³

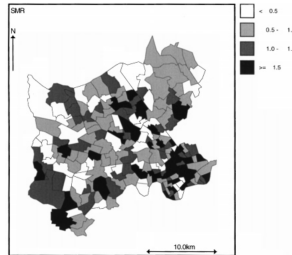


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What is epidemiological data ?

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- **Stratified** across time, space, and/or demographic groups ³
- Rare events : fraction of population with disease is much smaller than fraction tested

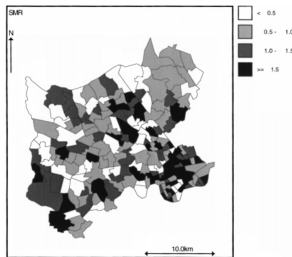


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- Commonly measured as **counts**, i.e. positive and no fractions
- **Stratified** across time, space, and/or demographic groups ³
- Rare events : fraction of population with disease is much smaller than fraction tested
- We like reporting in terms of rates and relative risks (in contrast to data we get)



Relative risk of larynx cancer mortality among men. From Pascutto et al. (2000)

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Goal : write down model for ρ_i , the **relative risk** in a region i

- Common to write the rate of occurrence as $\lambda_i = E\rho_i$, where E is the average rate across all regions
- ρ_i is a function of some covariates, i.e. tobacco sales
- Allow unmeasured variability between sites with **random effect** ε_i

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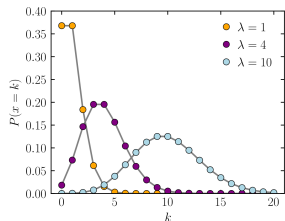
Generalized Mixed-effects Model ⁴

$$y_i \sim \text{Poisson}(\lambda_i)$$

$$\lambda_i = E\rho_i$$

$$\log \rho_i = \beta_0 + \beta_1 \text{tobacco}_i + \varepsilon_i$$

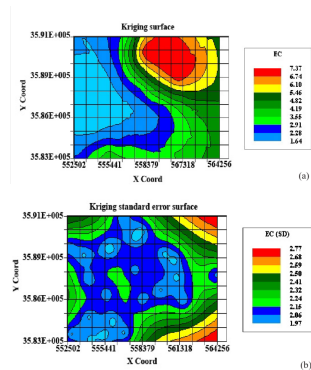
$$\varepsilon_i \sim N(0, \sigma_e^2)$$



wikipedia.org/wiki/Poisson_distribution

4. HARRISON et al., "A brief introduction to mixed effects modelling and multi-model inference in ecology", *PeerJ*. 2018.

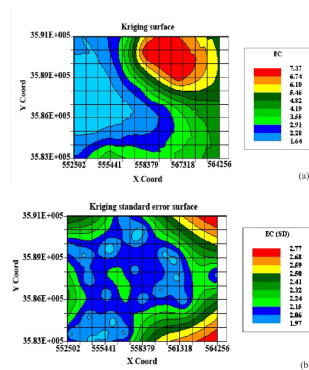
Spatial phenomena Z , measured at locations $(z_1, \dots, z_n)$



Kriging from classical geostatistics. From Amin et al. (2010).

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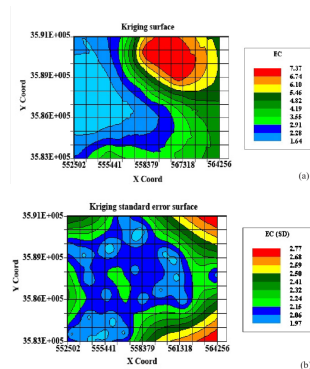
- “Nearby” spatial units similar
- Units with fewer “neighbors” have greater uncertainty



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Spatial phenomena Z , measured at locations $(z_1, \dots, z_n)$

- “Nearby” spatial units similar
- Units with fewer “neighbors” have greater uncertainty
- **Model-based geostatistics** : include z_i as a random effect in a hierarchical model!⁵



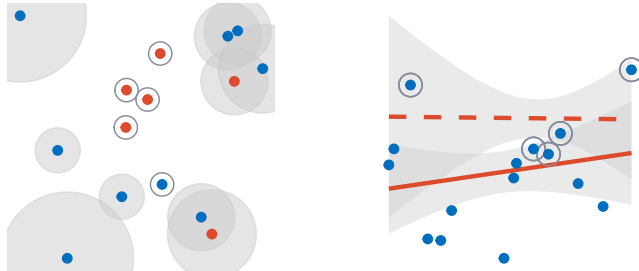
Kriging from classical geostatistics. From Amin et al. (2010).

5. DIGGLE, TAWN et MOYEED, “Model-based geostatistics”, *Journal of the Royal Statistical Society : Series C (Applied Statistics)*. 1998.

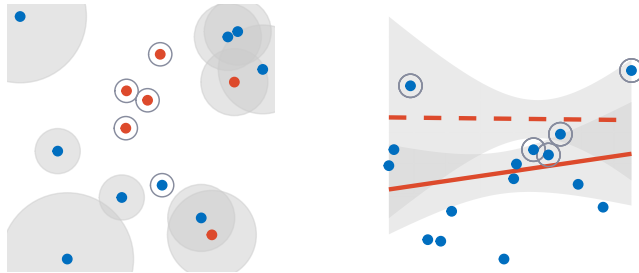
III

APPLICATION : ADAPTIVE SAMPLING AND SPATIAL STATISTICS FOR MANAGING CHAGAS DISEASE VECTORS

- Tradeoff between sampling to mitigate risk and sampling for inference
 - ▶ Leads to *selection bias*, which makes diagnosing risk less accurate



- Tradeoff between sampling to mitigate risk and sampling for inference
 - ▶ Leads to *selection bias*, which makes diagnosing risk less accurate
- A problem of *exploration* vs. *exploitation* in **design space** ⁶
 - (i) **Targeted control** : less exploration than random
 - (ii) **Targeted surveillance** : more exploration than random



6. NICHOLS et al., “Strategic testing approaches for targeted disease monitoring can be used to inform pandemic decision-making”, *PLOS Biology*. 2021.

- Chronic domestic exposure to Triatomines is primary route of infection

7. Aiga et al., "Chagas disease : assessing the existence of a threshold for bug infestation rate", *American Journal of Tropical Medicine and Hygiene*. 2012.

8. Yamagata and Nakagawa, "Control of Chagas disease", *Advances in Parasitology*. 2006.

9. Peterson et al., "Chagas disease in Central America : recent findings and current challenges in vector ecology and control", *Current Tropical Medicine Reports*. 2019

10. King et al., "Triatoma dimidiata infestation in Chagas disease endemic regions of Guatemala : comparison of random and targeted cross-sectional surveys", *PLoS Neglected Tropical Diseases*. 2011.

- Chronic domestic exposure to Triatomines is primary route of infection
- Keeping community infestation low can eliminate *T. cruzi* seroprevalence ⁷
 - ▶ Infestation prevalence below 5% a common target ⁸

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- Chronic domestic exposure to Triatomines is primary route of infection
- Keeping community infestation low can eliminate *T. cruzi* seroprevalence⁷
 - ▶ Infestation prevalence below 5% a common target⁸
- Complicated by limited resources/personnel, and by variability in vector ecology, socioeconomic risk factors, and cultural practices^{9, 10}
 - ▶ Efficient and localized strategies are required

7. Aiga et al., "Chagas disease : assessing the existence of a threshold for bug infestation rate", *American Journal of Tropical Medicine and Hygiene*. 2012.

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Targeted sampling could help in two key areas :

- (i) **Control.** Identify infested houses for treatment
- (ii) **Surveillance.** Assess 5% threshold throughout the community

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Take-home messages

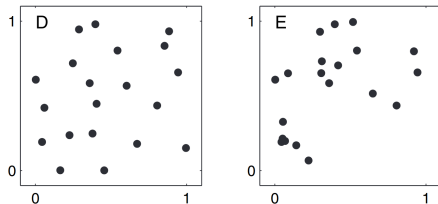
- These two goals are at odds
- Balancing these two goals helps jointly reduce the number houses checked/treated while still meeting the 5% reduction target
- Selecting houses to visit is an [experimental design](#) problem

- Given spatial phenomena Z over domain $\mathcal{D}$, choose discrete set $\mathcal{S} \subset \mathcal{D}$ to minimize variance over $\mathcal{D} \setminus \mathcal{S}$
 - ▶ Maximum or average variance can be used

11. MÜLLER et al., "Efficient prediction designs for random fields", *Applied Stochastic Models in Business and Industry*. 2015.

12. DIGGLE et LOPHAVEN, "Bayesian geostatistical design", *Scandinavian Journal of Statistics*. 2006.

- Given spatial phenomena Z over domain $\mathcal{D}$, choose discrete set $\mathcal{S} \subset \mathcal{D}$ to minimize variance over $\mathcal{D} \setminus \mathcal{S}$
 - ▶ Maximum or average variance can be used
- Tradeoff between prediction and inference of spatial parameters¹¹
 - ▶ Naturally accounted for in Bayesian paradigm¹²



Classical space-filling design (left) vs. Bayesian approach (right), for minimizing spatially-averaged variance. From Diggle and Lophaven (2006)

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Idea : use data *as it is gathered* to choose where to sample next

ADAPTIVE SAMPLING ALGORITHM

1. Sample initial design $\mathcal{S}_1$ uniformly and set $i = 1$
2. Fit the posterior distribution $p(\boldsymbol{\theta} \mid \mathbf{y}_i)$
3. Assign each location $\mathbf{s}_0 \in \mathcal{D} \setminus \mathcal{S}_i$ a utility $U(\mathbf{s}_0)$
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5. Set $i = i + 1$ and repeat steps 2-5 until done.

11. CHIPETA et al., "Adaptive geostatistical design and analysis for prevalence surveys", *Spatial Statistics*. 2016.

12. KABAGHE et al., "Adaptive geostatistical sampling enables efficient identification of malaria hotspots in repeated cross-sectional surveys in rural Malawi", *PLoS ONE*. 2017.

Idea : use data *as it is gathered* to choose where to sample next

- Initially proposed for spatial statistics by Chipeta et al. (2016)¹³ for Malaria surveys¹⁴
- Easier to find high-quality designs
- Adaptability is flexibility : makes it well-suited for surveys in public health

ADAPTIVE SAMPLING ALGORITHM

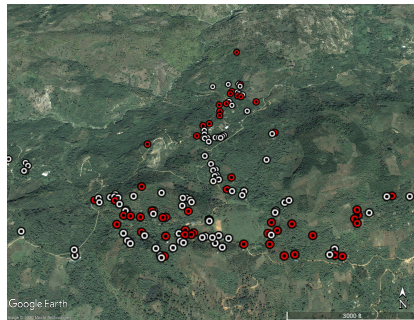
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Overview

- Data were collected from five villages in Chiquimula, Guatemala from August-December 2011^{15, 16}

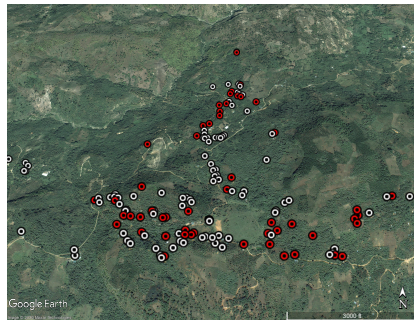


15. BUSTAMANTE et al., "Information to act : household characteristics are predictors of domestic infestation with the Chagas vector *Triatoma dimidiata* in Central America", *American Journal of Tropical Medicine and Hygiene*. 2015.

16. LIMA-CORDÓN et al., "Implementation science : epidemiology and feeding profiles of the Chagas vector *Triatoma dimidiata* prior to EcoHealth intervention for three locations in Central America", *PLoS Neglected Tropical Diseases*. 2018.

Overview

- Data were collected from five villages in Chiquimula, Guatemala from August-December 2011^{15, 16}
- Socioeconomic variables included number/type of domestic animals, housing materials, clutter around the home



15. BUSTAMANTE et al., "Information to act : household characteristics are predictors of domestic infestation with the Chagas vector *Triatoma dimidiata* in Central America", *American Journal of Tropical Medicine and Hygiene*. 2015.

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Logistic regression model

For each house $\mathbf{s} \in \mathcal{D}$, assume infestation status $y(\mathbf{s})$ follows a logistic regression model with spatial and non-spatial random effects :

$$y(\mathbf{s}) \sim \text{Bernoulli}(r(\mathbf{s}))$$
$$\text{logit}(r(\mathbf{s})) = \eta(\mathbf{s}) = \boldsymbol{\beta}^\top \mathbf{x}(\mathbf{s}) + z(\mathbf{s}) + \varepsilon(\mathbf{s}),$$

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$$\text{logit}(r(s)) = \eta(s) = \beta^\top x(s) + z(s) + \varepsilon(s),$$

r : "risk" or probability of infestation

η : Gaussian linear predictor

x : measured covariates

β : fixed effect coefficients

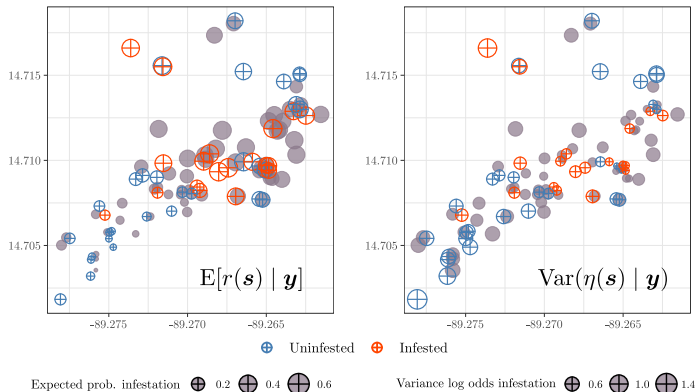
z : residual spatial variation

ε : residual non-spatial variation

Posterior prediction

Suppose we have infestation data from a subset of houses, $\mathbf{y} = \{y(\mathbf{s}) : \mathbf{s} \in \mathcal{S}\}$

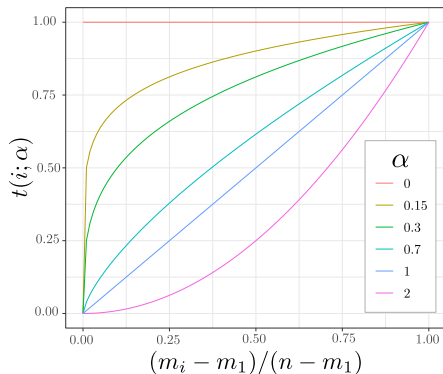
▷ The probability a house is infested is $\Pr(\hat{y}(\mathbf{s}) = 1 \mid \mathbf{y}) = \mathbf{E}[\hat{r}(\mathbf{s}) \mid \mathbf{y}]$



- **Sample infested locations**
 - ▷ use normalized risk $\tilde{r}(s_0)$
- **Minimize sampling bias**
 - ▷ use normalized variance $\tilde{v}(s_0)$

$$U(\mathbf{s}_0) = t(i; \alpha) \tilde{r}(\mathbf{s}_0) + (1 - t(i; \alpha)) \tilde{\nu}(\mathbf{s}_0)$$

- **Sample infested locations**
 - ▷ use normalized risk $\tilde{r}(\mathbf{s}_0)$
- **Minimize sampling bias**
 - ▷ use normalized variance $\tilde{\nu}(\mathbf{s}_0)$
- Interpolate with **transition function** $t(i; \alpha)$



To assess whether 5% infestation target is met, we are interested in state of whole village, not just individual homes

- Need the joint distribution of unvisited houses, $\Pr(\hat{\mathbf{y}}_0 \mid \mathbf{y})$

	$\hat{y}_1$	$\hat{y}_2$	$\hat{y}_3$	$\cdots$
$\hat{\mathbf{y}}_0^{(1)}$	1	0	0	$\cdots$
$\hat{\mathbf{y}}_0^{(2)}$	1	1	1	$\cdots$
$\hat{\mathbf{y}}_0^{(1)}$	0	1	1	$\cdots$
$\vdots$		$\vdots$		$\ddots$

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$\hat{\mathbf{y}}_0^{(2)}$	1	1	1	$\cdots$
$\hat{\mathbf{y}}_0^{(1)}$	0	1	1	$\cdots$
$\vdots$		$\vdots$		$\ddots$

- Use to approximate the **termination condition**

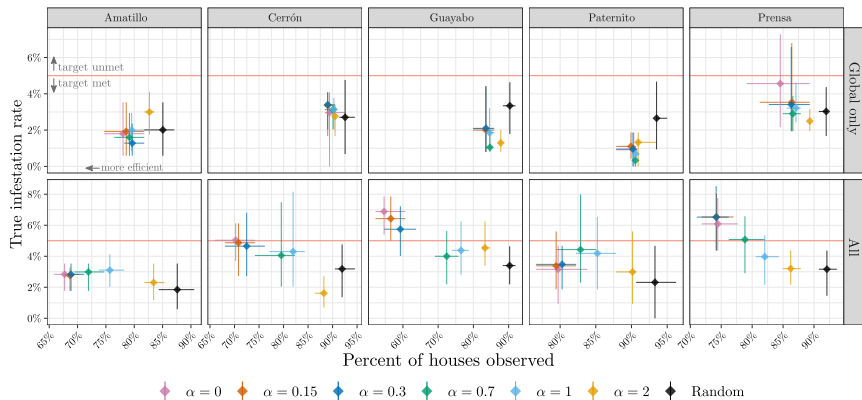
$$\Pr(\mathbf{1} \cdot \hat{\mathbf{y}}_0 < \kappa n) \geq \gamma,$$

for **reduction target** κ and **confidence level** γ

Experimental setup

- Two situations of available covariates :
 - (i) **Global only** : only remote information
 - (ii) **All** : remote + socioeconomic information
- For each village, select 50 random initial designs of 10 houses
- For each initial design, compare range of $\alpha = 0$ (most greedy) to $\alpha = 2$ (most exploratory) with random sampling
- When termination condition met, record number of houses visited and percentage of remaining infestations

Results



▷ Adaptive sampling always more efficient, but additional covariate information leads to higher bias when α too low

Chagas paper

Case BKM, Young JG, Penados D, Monroy C, Hébert-Dufresne L, Stevens L. "Spatial epidemiology and adaptive targeted sampling to manage the Chagas disease vector *Triatoma dimidiata*", *PLOS Neglected Tropical Diseases*. 16(6) : e0010436.
doi.org/10.1371/journal.pntd.0010436.

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Funding and computational resources

